# Answer key: Lesson 10, Inquirers' Showcase

**For the tutor and grown-ups.** Keep this page away from the child until they have finished and checked their own work. Ask *how do you know?* before saying right or wrong.

## Start here (pages 1 and 2)

**S1.** Look back at your year books. Choose a favourite page, picture or puzzle. Many answers are right.

**Answer: Many answers are right**

Many answers are right. A good answer names the page, picture or puzzle and gives one reason.

**Watch for:** Writing only that it was fun: add one thing about the maths.; Leaving it blank: you may pass.

**S2.** Tick one idea in each row. Nobody sees your ticks except your grown-up and your tutor.

**Answer: Many answers are right**

Many answers are right. The ticks show what to practise: an idea you are learning is a good idea to teach a toy.

**Watch for:** Ticking one box for every idea without thinking.; Feeling that one box is better: neither box is better.

**S3.** Which one is a clear puzzle? Write its number in the brackets.

**Answer: (2)**

Option (2) has a story (Ali and Mei), numbers (6 and 3) and one question. Option (1) has no numbers. Option (3) has no question. Option (4) has no story or numbers.

**Watch for:** Choosing (4), a question with no story.; Choosing (3), a story with no question.

**S4.** A child in another class wrote a puzzle and an answer. Siti has 9 stickers. Ravi has 4 more stickers than Siti. How many stickers does Ravi have? The child says 5.

**Answer: Wrong; 13 stickers**

Ravi has 4 more than 9, so 9 + 4 = 13 stickers. The answer 5 is 9 take away 4, which is not what 4 more means. Check: 13 take away 9 is 4.

**Watch for:** Saying Right because 9 and 4 are in the puzzle.; Writing 5 (taking away).

**S5.** Mi makes a ten first to add 9 + 7. Fill in the blanks.

**Answer: 1; 6; 16**

9 and 1 make 10. 7 take away 1 is 6. 10 and 6 make 16.

**Watch for:** Writing 2 for the first blank.; Forgetting to take the 1 from the 7.

**S6.** Count by rows. Look at the dots.

**Answer: 4 rows; 20 dots**

There are 4 rows of 5 dots: 5, 10, 15, 20. There are 20 dots in all.

**Watch for:** Writing the dots in one row only.; Writing 9 (4 and 5 added).

## Think harder (page 3)

**T1.** Write a puzzle for a new Level 1 child. It needs a story, numbers, one question and one answer. Use two numbers up to 20. Many answers are right.

**Answer: Many answers are right**

Many answers are right. A good puzzle has a story, numbers, one question and exactly one answer, for example: Wei has 8 stickers. Lily has 5 more stickers than Wei. How many stickers does Lily have? The answer is 13.

**Watch for:** Writing a story with no question.; Writing a puzzle with two answers.

**T2.** Read your puzzle from question T1 to your quest partner. Your partner solves it. Fill in the blanks.

**Answer: Many answers are right**

Many answers are right. If your partner gets a different answer, your puzzle may have two answers or a mistake. Check it again.

**Watch for:** Telling your partner the answer first.; Not checking a different answer.

**T3.** Puzzle A: Write a number bigger than 10 and smaller than 15. Puzzle B: I am a number. I am 4 more than 9. Puzzle C: Ali has 3 pens and Mei has 4 pens.

**Answer: B; 4**

Puzzle B has exactly one answer: 13. Puzzle A has four answers: 11, 12, 13 and 14. Puzzle C has no question.

**Watch for:** Choosing C (it has numbers but no question).; Writing 5 for puzzle A (11 to 15 counted).

## Go further (page 4)

**G1.** Mi's puzzle: Two numbers add up to 12. One number is 2 more than the other. Fill in the table of every pair that makes 12. Then answer (a) and (b).

**Answer: 5 and 7; 1 pair**

The pairs are 0 and 12, 1 and 11, 2 and 10, 3 and 9, 4 and 8, 5 and 7, 6 and 6. The differences are 12, 10, 8, 6, 4, 2 and 0. Only 5 and 7 differ by 2. So the numbers are 5 and 7, and no other pair works.

**Watch for:** Stopping at 5 and 7 without listing every pair.; Writing 6 and 6 (the difference is 0).

**G2.** A child in another class wrote a puzzle: Two numbers add up to 10. One number is 4 more than the other. The numbers are 6 and 4.

**Answer: No; 2; 3 and 7**

6 + 4 = 10, so the first clue fits. 6 take away 4 is 2, not 4, so the second clue does not fit. The numbers that fit are 3 and 7: 3 + 7 = 10 and 7 take away 3 is 4.

**Watch for:** Saying Yes because the numbers add up to 10.; Writing 4 for the difference.

**G3.** Write a puzzle with a rule for a friend. Solve it. Then show why your answer must be right: draw it, list every case, or check a second way. Many answers are right.

**Answer: Many answers are right**

Many answers are right. A good puzzle has a rule, one answer, and a proof: every case is listed, or the answer is checked a second way. For example: Two numbers add up to 8. One is 2 more than the other. 0 and 8, 1 and 7, 2 and 6, 3 and 5, 4 and 4. Only 3 and 5 are 2 apart.

**Watch for:** Writing a puzzle with two answers.; Copying a puzzle from a book or from class.

## Investigation (page 5): Teach it in a minute

**Many answers are right. A good plan names the idea, gives an example with numbers or a picture, and one check.**

**Look for:** A child who names an idea from the year, gives one example with numbers or a picture, and says one check. Credit any clear idea. A child who writes only the name has the first step: ask, what is an example? A child who chooses an idea from the last lessons has looked back well; one from Term 1 has too.

## Exit ticket (page 5)

**E1.** Mei thinks of a number. She takes away 5 and gets 9.

**Answer: 14**

Work backwards: 9 and 5 make 14. Check: 14 take away 5 is 9.

**Watch for:** Writing 4 (9 take away 5).; Not checking.

**E2.** Which puzzle has exactly one answer? Write its number in the bracket.

**Answer: (2) 11**

Option (2) has one answer: 3 more than 8 is 11. Option (1) has many answers. Option (3) has no question. Option (4) has no story.

**Watch for:** Choosing (1): many numbers are bigger than 6.; Choosing (3): no question.

**E3.** A child in another class says 7 + 8 = 16.

**Answer: Wrong**

7 + 8 = 15, not 16. Check: 16 take away 8 is 8, not 7.

**Watch for:** Choosing Right without checking.; Writing 17.

## Sneak-peek puzzles (not in the workbook)

1. Many questions are right: How many pens in all? (10) How many more pens does Ali have than Mei? (2).
2. No. Many numbers fit (4, 5, 6 and 7). A good puzzle has exactly one answer.